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Deep learning and GIS integration for plant disease diagnosis and management

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Abstract

Plant diseases pose a major threat to global food security, particularly in agricultural regions such as Egypt. This study presents an integrated framework that combines deep learning (DL) with Geographic Information Systems (GIS) for the automated diagnosis and spatial monitoring of tomato and potato diseases. Three deep learning models were developed and evaluated: DenseNet169, MobileNetV2, and a Custom-CNN baseline. Comparative experiments on the test dataset showed that DenseNet169 achieved the highest evaluation accuracy of 98.24%, followed by MobileNetV2 with 94.66%, while the Custom-CNN obtained a lower accuracy of 82.60%, reflecting the limitations of shallow architectures trained from scratch. Owing to its superior performance, DenseNet169 was selected for integration with the GIS module. The GIS component geotags disease predictions and visualizes outbreaks through interactive crop-specific layers, heatmaps, and temporal tracking, enabling early detection and targeted interventions. This spatial–temporal integration allows stakeholders to monitor disease progression, assess risk under varying environmental conditions, and optimize resource allocation for precision agriculture. The final system also incorporates a bilingual web interface (Arabic/English) enhanced with Grad-CAM visualizations, which improve interpretability and build trust among local farming communities. Field validation conducted in Beheira Governorate, Egypt, confirmed the accuracy and practical utility of the system in real-world agricultural settings. By integrating AI-based diagnostics with geospatial intelligence, the proposed framework provides a scalable and sustainable decision-support tool, offering a valuable pathway toward data-driven plant disease management in developing regions.

Keywords Plant disease, Deep learning, CNN, GIS, MobileNetV2, Disease mapping, Egypt, Smart agriculture

1 Introduction

Agriculture represents a fundamental pillar of the Egyptian economy, accounting for approximately 11.3% of the national Gross Domestic Product (GDP) and employing nearly 28% of the labor force. The productivity and quality of agricultural crops are highly susceptible to climatic and environmental variations, which in turn influence economic stability and the country's competitiveness in international markets. On a global



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scale, plant-based products constitute over 80% of the human diet, underscoring the central role of crop health in ensuring food security. However, plant diseases—originating from bacterial, fungal, or viral pathogens—continue to pose a significant threat to agricultural output. According to the Food and Agriculture Organization (FAO), plant pests and diseases are responsible for an estimated 20–40% of annual crop losses worldwide, thereby presenting a critical obstacle to the sustainability of food systems [1]. In the Egyptian context, enhancing plant disease management is imperative for achieving both sustainable food security and economic resilience.

Timely and accurate diagnosis of plant diseases is essential to minimizing crop damage, as early detection enables targeted treatment before the disease spreads extensively. Among the various indicators of plant health, leaf symptoms remain the most visible and informative, making leaf image analysis a practical and accessible diagnostic tool. Despite this, Egypt suffers from a noticeable shortage of agricultural extension personnel, limiting farmers' access to expert guidance. Consequently, many farmers resort to self-diagnosis or depend on unverified online resources, which often leads to inaccurate identification of diseases and the implementation of inappropriate management strategies [2].

Recent advancements in Artificial Intelligence (AI), particularly in deep learning, have opened new frontiers in the development of automated plant disease detection systems. The proliferation of smartphones and the accessibility of GPU-powered computing have further facilitated the deployment of such technologies in real-world agricultural contexts. Among various AI techniques, Convolutional Neural Networks (CNNs) have exhibited remarkable efficacy in image-based classification tasks, including plant disease identification. Several studies have demonstrated the successful application of CNN architectures such as VGG16, InceptionV3, DenseNet201, and MobileNetV2, achieving high classification accuracy while maintaining computational efficiency [3, 4].

Concurrently, GIS have gained prominence as a powerful tool for analyzing, managing, and visualizing spatially referenced data. GIS integrates hardware, software, and analytical methodologies to support the geospatial representation of phenomena, including the spread of plant diseases [5]. By enabling the construction of spatial distribution maps, GIS contributes to identifying disease hot-spots, monitoring crop health dynamics, and informing precision agriculture practices [6]. The integration of GIS with real-time plant disease diagnostics offers the potential to support early warning systems and enhance evidence-based agricultural planning and interventions.

Despite these parallel advancements, current research efforts often treat deep learning-based diagnosis and GIS applications as distinct domains. Most AI-driven plant disease detection models are limited to classifying disease symptoms from images, lacking spatial contextualization or actionable insights for decision support. Conversely, GIS applications in agriculture frequently rely on manually reported data and rarely incorporate automated, real-time diagnostic outputs. This gap underscores the need for an integrated approach that combines deep learning and GIS to enhance both the accuracy of disease detection and the spatial intelligence of agricultural decision-making.

1.1 Research gap and motivation

Most existing deep learning-based plant disease diagnosis systems are limited to static image classification, overlooking the spatial and temporal aspects of disease occurrence.

This shortcoming prevents stakeholders from understanding not only where diseases appear but also how they evolve and spread over time, thereby constraining opportunities for timely intervention and strategic disease management.

To overcome this limitation, the present study aims to integrate deep learning–based disease diagnosis with GIS, enabling spatial visualization and dynamic monitoring of plant health. This integration bridges the gap between image-level recognition and field-scale management, fostering precision agriculture and proactive decision-making.

The motivation for this research is both scientific and practical. In countries like Egypt—where agriculture is central to food security and rural livelihoods—timely and accurate diagnosis of crop diseases can minimize losses and reduce excessive pesticide use. Moreover, by deploying lightweight deep learning models connected to GIS visualization tools, the framework offers an accessible and cost-effective solution that empowers smallholder farmers and agricultural officers with real-time, actionable insights.

The major contributions of this research can be summarized as follows. First, among the evaluated deep learning architectures, DenseNet169 achieved the highest classification accuracy of 98.24% across 13 tomato and potato disease classes from the PlantVillage datasets, confirming its robustness and generalization ability. Second, this study introduces a novel integration of CNNs and GIS that enables real-time geospatial visualization and spatio-temporal monitoring of plant disease outbreaks, thereby advancing beyond existing frameworks that focus primarily on static visualization or image-based diagnosis, such as DeepAgroVis, SmartFarmGIS, and the recently proposed AgroDetect [7]. Unlike these systems, which primarily focus on individual plant-level analysis, the proposed framework extends to dynamic, large-scale monitoring by transforming model outputs into interactive GIS layers for real-time disease tracking and management. Third, a predictive disease spread estimation component is implemented to correlate environmental and geographical factors with disease occurrence density, offering early-warning capabilities for agricultural stakeholders. Fourth, a bilingual web-based decision-support platform has been developed to ensure accessibility for both local farmers and researchers. Furthermore, model interpretability is enhanced using Grad-CAM visualization, which highlights the most relevant regions of infected leaves to promote transparency and explainable in diagnostic decisions. Finally, field validation conducted in El-Beheira governorate, Egypt, confirmed the framework's practical utility and adaptability to real-world agricultural environments, establishing it as a comprehensive precision agriculture tool that unites automated diagnosis, spatial analytics, and decision support.

This paper is organized as follows: Sect. 2 reviews related work on deep learning and GIS for plant disease diagnosis. Section 3 describes the methodology and evaluation of DenseNet169, MobileNetV2, and Custom-CNN. Section 4 outlines the experimental setup, while Sect. 5 presents the results and GIS integration. Finally, Sects. 6–8 discuss limitations, future directions, and conclusions.

2 Literature review

2.1 Deep learning in plant disease diagnosis

Deep learning has revolutionized plant disease diagnosis by enabling automated, image-based recognition of visual symptoms on leaves. Over the past decade, CNNs have emerged as the dominant architecture due to their hierarchical feature extraction

capabilities and high classification accuracy. Existing research on CNN-based diagnosis can be conceptually grouped into three major themes: (1) model architecture and transfer learning approaches, (2) hybrid and optimized network designs, and (3) lightweight and mobile-oriented architectures for field deployment.

(1) Model Architecture and Transfer Learning Approaches.

Numerous studies have demonstrated the effectiveness of pretrained CNNs for plant disease recognition using benchmark datasets such as *PlantVillage*. For instance [8], achieved 98.75% test accuracy, while [9] compared VGG16, InceptionV3, ResNet152V2, and DenseNet201, identifying DenseNet201 as the top performer (99.30% validation accuracy). Similarly, Too [10] reported that DenseNet121 outperformed other architectures with 99.75% accuracy on 54,306 images, reinforcing the effectiveness of deep transfer learning in plant pathology.

(2) Hybrid and Optimized Network Designs.

Recent research has advanced beyond conventional CNNs toward hybrid and ensemble architectures that enhance feature representation and robustness. In [11], a Region-based Convolutional Neural Network–Region-based Fully Convolutional Network (RCNN–RFCN) hybrid model achieved 98.5% accuracy, while [12] proposed UnitedModel, a combined CNN framework that reached 99.17% accuracy for grape leaf disease detection. These hybridization strategies reflect a growing trend toward model fusion for improved generalization and contextual understanding.

(3) Lightweight and Mobile-Oriented Architectures.

Given the demand for real-time diagnosis in agricultural environments, lightweight CNNs have gained increasing attention. In [13] employed depthwise separable convolutions to reduce computational cost while maintaining 99.56% accuracy across 38 diseases, making the model suitable for mobile deployment. Similarly [14], found that ShuffleNet achieved a reasonable 83.68% accuracy with significantly fewer parameters, illustrating the balance between model complexity and efficiency.

Despite these advancements, several studies in [15] highlight ongoing challenges related to dataset variability, background noise, and environmental conditions that hinder model generalization beyond controlled datasets such as *PlantVillage*. Therefore, integrating deep learning with contextual and geospatial tools like GIS represents a crucial step toward developing field-ready, intelligent decision-support systems for sustainable agriculture.

2.2 GIS in plant disease monitoring and management

GIS and Remote Sensing (RS) technologies have become indispensable tools in modern plant disease monitoring and management, providing spatial and temporal insights that enhance early detection, forecasting, and intervention. Existing research in this area can be grouped into three major thematic domains: (1) environmental and epidemiological mapping, (2) precision agriculture and decision-support systems, and (3) geospatial forecasting and sustainability applications.

(1) Environmental and Epidemiological Mapping.

Several studies have demonstrated how GIS and RS facilitate the visualization and analysis of environmental factors driving plant disease dynamics. In [16] integrated GIS and RS to map disease–environment interactions across multiple crops, enabling early

detection of wheat yellow rust, rice sheath blight, and potato late blight. Similarly [17], employed high-resolution satellite imagery and Unmanned Aerial Vehicle (UAV) data to assess spatial variability in crop health, illustrating how geospatial layers can reveal early symptoms before visible damage occurs.

(2) Precision Agriculture and Decision-Support Systems.

The use of GIS-based mobile and web platforms for real-time agricultural monitoring has expanded rapidly [18]. developed an Android-based GIS application for pest and disease monitoring, demonstrating strong reliability and user adoption [19]. reviewed GIS–RS advancements in oil palm management, emphasizing the integration of machine learning and deep learning to optimize precision agriculture workflows. Likewise [20], demonstrated the use of GIS, Global Navigation Satellite System (GNSS), and hyperspectral remote sensing within precision agriculture frameworks for targeted disease management, achieving reduced pesticide application and improved decision efficiency.

(3) Geospatial Forecasting and Sustainable Management.

Forecasting models using GIS have proven valuable for proactive disease control and sustainable planning [21, 22]. highlighted the role of geospatial forecasting in developing disease-resistant crop strategies and timely interventions. Previous studies demonstrated how GIS–RS integration supports early detection and large-scale surveillance, while [23, 24] emphasized its capacity to assess both biotic and abiotic stressors influencing plant health and productivity.

Collectively, these studies confirm GIS and RS as vital components of plant disease management ecosystems. However, most implementations remain crop-specific, regional, and detached from AI-driven diagnosis systems. This gap underscores the need for integrated DL–GIS frameworks capable of combining image-based diagnosis with spatial and environmental intelligence to enable predictive, data-informed agricultural decision-making.

2.3 Integration of deep learning and GIS in agriculture

While DL demonstrates strong capabilities in classifying plant diseases from leaf images and GIS excel in spatial analysis, their integration in real-time agricultural decision-support systems remains limited. DL models typically lack spatial and temporal context, whereas GIS often depends on manually collected or delayed data. By combining the two, static image-based diagnoses can be transformed into dynamic, location-aware insights that enable visualization of disease hotspots, detection of temporal trends, and more efficient resource allocation.

Although some recent studies, such as Suresh et al. [7], have applied deep learning (DL) for plant disease monitoring, most approaches remain experimental or limited in scope, often lacking real-time functionality or large-scale validation. This underscores a critical research gap, which the present study addresses by developing a unified DL–GIS framework specifically designed for the diagnosis, mapping, and management of tomato and potato diseases.

2.4 Summary and research gap

The reviewed literature confirms that CNN-based deep learning models achieve high diagnostic accuracy, while GIS and remote sensing play critical roles in spatial analysis and disease monitoring. However, these two domains have largely evolved

independently, limiting the delivery of real-time, spatially aware decision-support systems. Deep learning models typically lack geospatial context, and GIS-based monitoring systems often depend on delayed or manually collected data.

To address this research gap, this study proposes an integrated deep learning–GIS framework for intelligent plant disease management. Three models—DenseNet169, MobileNetV2, and a Custom-CNN—were comparatively evaluated, with DenseNet169 demonstrating the best performance (98.24% accuracy). Accordingly, DenseNet169 was selected for GIS integration to visualize disease hotspots and analyze potential spread dynamics using environmental parameters.

This integration enables automatic mapping of diagnosed cases, temporal tracking of outbreaks, and early identification of high-risk zones. By linking diagnostic intelligence with geospatial analytics, the proposed system contributes a practical, scalable, and sustainable solution to enhance agricultural resilience and support data-driven disease management.

3 Methodology

3.1 Overview of the proposed model

In this study, three deep learning models were developed to classify healthy and diseased tomato and potato leaves using the PlantVillage dataset with a 70/15/15 split for training, validation, and testing. The models included two transfer learning architectures—DenseNet169 and MobileNetV2—alongside a Custom-CNN built from scratch. Experimental results showed that DenseNet169 achieved the highest accuracy (98.24%), followed by MobileNetV2 (94.66%), while the Custom-CNN recorded a lower accuracy (82.60%) due to its shallow depth and limited feature extraction capability.

Based on these results, DenseNet169 was selected for integration with the GIS module, as it provided superior accuracy and robustness, making it suitable for precise and reliable deployment.

The proposed framework integrates DenseNet169-based diagnosis with GIS, enabling both automated classification and spatial visualization of disease outbreaks. CNN architectures effectively capture hierarchical features from leaf images, while transfer learning accelerates convergence and improves generalization. The overall pipeline combines pre-processing, prediction, and geospatial mapping to deliver a scalable and practical decision-support system for agriculture (Fig. 1).

Recent works have emphasized the value of transfer learning and spatially enabled deep-learning frameworks in agricultural image analysis. For example, In [25] proposed an automated annotation and segmentation framework for multi-date aerial image analysis, demonstrating how deep neural networks can extract spatially meaningful patterns from agricultural imagery. Inspired by this line of research, the present study adopts a similar deep-learning-driven approach for spatially referenced disease detection, integrating CNN-based diagnosis with GIS visualization to enhance interpretability and decision support.

3.2 Tomato and potato leaf disease dataset

This study utilized a curated subset of the PlantVillage dataset comprising 18,163 high-resolution RGB images of tomato and potato leaves across 13 classes (9 for tomato and 4 for potato), covering both healthy and diseased samples such as Early Blight, Late Blight,

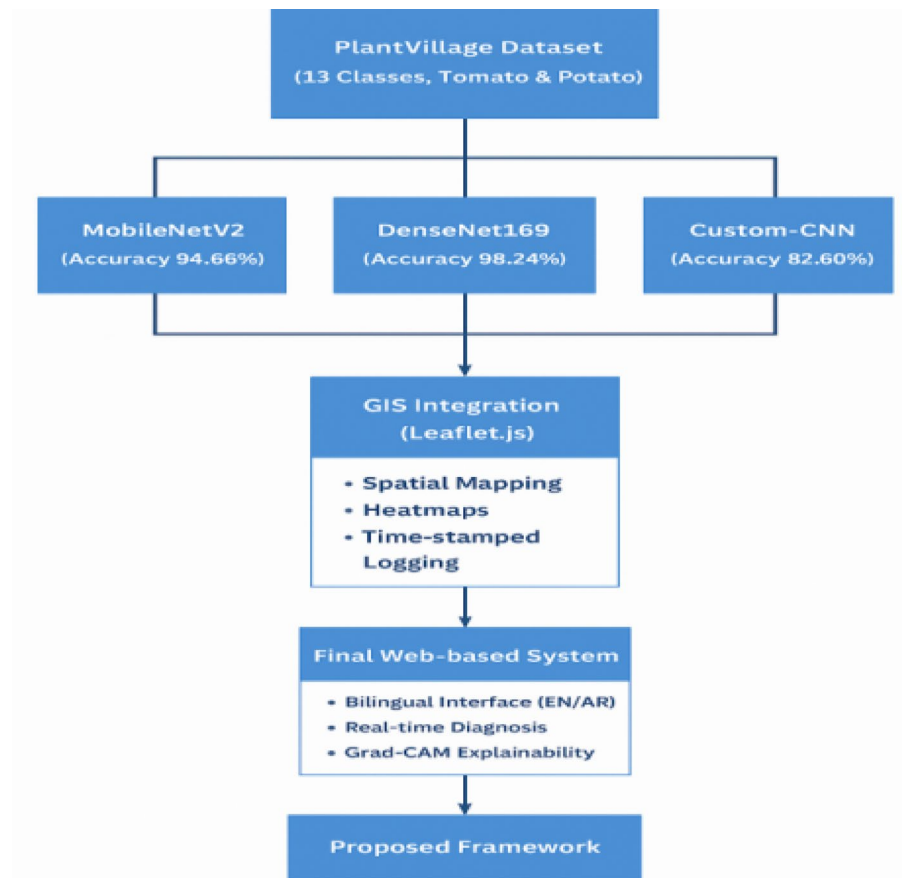


Fig. 1 Overall workflow of the proposed system framework

Septoria Leaf Spot, Yellow Leaf Curl Virus, and bacterial infections. To ensure balanced representation, stratified sampling was applied. The dataset was divided into 70% for training, 15% for validation, and 15% for testing.

To provide a more intuitive overview of the dataset, Fig. 2 presents representative samples from the 13 tomato and potato classes. This visual summary highlights the diversity of disease symptoms and healthy leaves, illustrating the variability that the models must learn to distinguish during classification.

3.3 Image preprocessing

To ensure consistency and optimize model performance, all images were resized from 256×256 to 224×224 pixels, aligning with the input requirements of the selected CNN-based architectures (DenseNet169, MobileNetV2, and a Custom-CNN). Pixel values were normalized to the range $[0, 1]$ by dividing each RGB channel by 255, which stabilized the learning process, standardized input distributions, and facilitated faster convergence. These preprocessing steps effectively conditioned the dataset for robust training and evaluation across the four models.

3.4 Augmentation process

To enhance CNN robustness and mitigate overfitting, data augmentation was applied using TensorFlow's ImageDataGenerator. Transformations included horizontal and vertical flips, random rotations ($\pm 20^\circ$), zooming ($\pm 20\%$), width and height shifts, shearing,

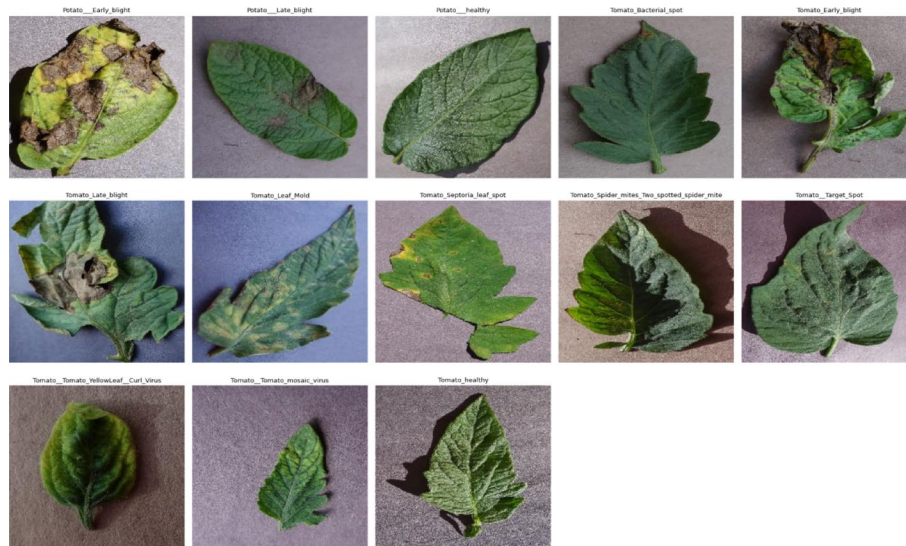


Fig. 2 Presents representative samples from the 13 tomato and potato classes

brightness adjustments, and rescaling, simulating natural variations in leaf orientation, size, and lighting conditions. Augmentation was applied consistently for DenseNet169, MobileNetV2, and a Custom-CNN, while a lighter augmentation pipeline was adopted for the custom CNN to reduce training complexity. This approach contributed to improved generalization and stability across all evaluated models.

3.5 Dataset partitioning and evaluation strategy

The dataset was partitioned using a stratified 70:15:15 split, resulting in 12,709 images for training, 2,719 for validation, and 2,735 for testing, with balanced representation across all 13 classes. This ensured fair evaluation and prevented data leakage between the training, validation, and testing sets. The same partitioning strategy was applied consistently across all three models: DenseNet169, MobileNetV2, and Custom-CNN.

Early stopping was implemented during training to prevent overfitting and improve generalization. Model performance was assessed using multiple metrics, including Accuracy, Precision, Recall, F1-score, and Support, complemented by confusion matrices for class-specific evaluation. Training and validation curves were monitored to track convergence, while ROC curves and AUC scores provided additional insight into each model's discriminative capability across disease classes.

This evaluation framework enabled a robust comparison of the three architectures, highlighting their relative trade-offs in terms of accuracy, efficiency, and practical deployment suitability.

3.6 Fine-tuning strategy

Fine-tuning was performed by optimizing key hyperparameters such as batch size, optimizer, learning rate, and validation strategy [10]. For DenseNet169 and MobileNetV2, training was conducted in two stages: 20 frozen epochs followed by 20 unfrozen epochs, allowing gradual adaptation of pre-trained weights. In contrast, the Custom-CNN was trained in a single stage of 40 epochs. Early stopping was applied consistently across all models to prevent overfitting and ensure fair comparison under unified experimental conditions.

3.7 Model comparison strategy

To ensure an objective and reproducible assessment of model performance, the three architectures—DenseNet169, MobileNetV2, and the Custom-CNN—were trained and evaluated under identical experimental conditions, using the same data splits, augmentation settings, and optimization parameters. The comparative analysis focused on accuracy, loss, and generalization ability across the test dataset.

To statistically validate whether the observed performance differences were significant, McNemar's test was applied for pairwise model comparisons. This non-parametric test evaluates the degree of agreement and disagreement between two classifiers on the same dataset, offering a robust framework for assessing comparative performance reliability [26]. This statistical evaluation approach aligns with recent methodological standards for comparative deep learning model validation, which emphasize the integration of significance testing to ensure that observed performance differences are not due to random variation [27].

3.8 Training the models

Three models were trained for tomato and potato disease classification: DenseNet169, MobileNetV2, and a Custom-CNN. The Custom-CNN was designed with three convolutional blocks followed by fully connected layers, while the transfer learning models (DenseNet169 and MobileNetV2) adopted a unified classification head during fine-tuning. All models were trained using the same dataset partitioning and evaluation strategy described earlier, ensuring consistency in comparison.

3.9 Feature extraction and Explain-ability

The three models adopted distinct feature extraction strategies. DenseNet169 utilized dense connectivity across layers, facilitating efficient feature reuse and enabling the extraction of highly discriminative representations for fine-grained disease classification. MobileNetV2 employed depth-wise separable convolutions and inverted residual blocks with linear bottlenecks, producing compact yet expressive feature maps that are particularly suitable for lightweight deployment in real-time systems. In contrast, the Custom-CNN, although less accurate, learned hierarchical features from scratch, progressing from low-level edges and textures to higher-level symptomatic leaf structures.

To enhance interpretability, Grad-CAM was applied to DenseNet169 and MobileNetV2, producing class-discriminative heatmaps that highlight the leaf regions most relevant to model predictions. These visualizations validated that the models consistently attend to disease-affected areas, thereby reinforcing their reliability and transparency in practical applications. Similar applications of Grad-CAM in plant pathology have demonstrated its effectiveness in confirming model focus on symptomatic regions in grape leaves [28] and in pest detection frameworks [29]. Representative examples from our study are presented in Fig. 3 (sample from the training dataset) and Fig. 4 (field sample from Badr Village, Beheira, January 2025), illustrating both accuracy and robustness under real conditions.

These visual comparisons validated the model's transferability and practical relevance in real agricultural environments.

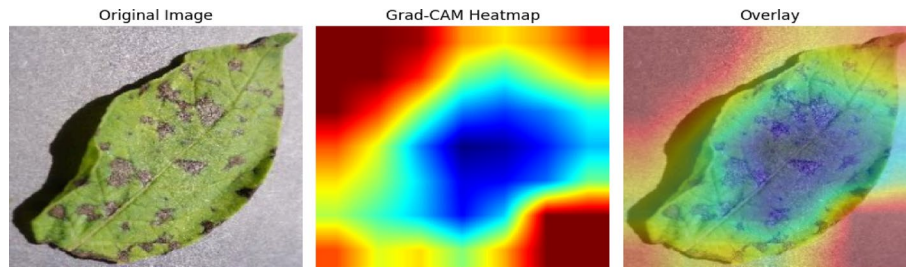


Fig. 3 An image from the PlantVillage dataset used during training

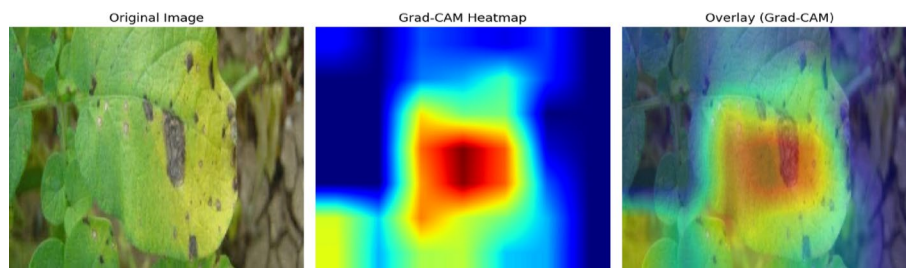


Fig. 4 A real-field image captured from Badr village, Beheira Governorate, Egypt

3.10 GIS-Driven disease monitoring and environmental integration

3.10.1 Role of GIS in plant disease management

Geographic Information Systems (GIS) provide a spatial framework for analyzing the distribution of plant diseases and their environmental determinants. Because disease outbreaks are strongly influenced by climatic and ecological factors, GIS facilitates early detection, continuous monitoring, and the design of targeted intervention strategies. Previous studies have demonstrated that integrating GIS with AI-based diagnostic models enhances the tracking and prediction of plant diseases, particularly for potato late blight and viral infections in tomato crops.

In this study, GIS was combined with deep learning models to map the spatial occurrence of tomato and potato diseases across different regions. This integration enabled not only accurate disease diagnosis at the leaf level but also the visualization of disease spread patterns at both field and regional scales. By linking diagnosis with geospatial insights, the framework provides stronger decision-support tools for farmers and policymakers, promoting rapid interventions to mitigate large-scale outbreaks.

3.10.2 GIS-Based visualization and Multi-Crop layering

The integration of DenseNet169 outputs into an interactive GIS platform was achieved using Leaflet.js, enabling real-time visualization of disease occurrences. Each diagnostic result was geotagged and displayed on the map as a color-coded marker: red for tomato diseases and brown for potato diseases. To enhance interpretability, the system supports multi-layer visualization, allowing users to toggle crop-specific layers independently. This functionality provides a clear spatial overview of disease dynamics across crops, facilitates comparative analysis, and supports targeted management interventions at both farm and regional levels.

3.10.3 Diagnosis logging and metadata recording

All disease predictions generated by DenseNet169 are automatically logged within the system together with detailed metadata, including the confidence score, detection timestamp, crop type, GPS coordinates, and key environmental parameters (e.g., temperature, humidity, and soil moisture). This structured logging framework ensures full traceability of diagnostic outputs and provides a reliable basis for historical trend analysis across different seasons and geographic regions. By capturing both spatial and temporal dimensions, the recorded data support advanced epidemiological studies, enabling the modeling of disease spread dynamics, identification of high-risk areas, and evaluation of intervention outcomes. Furthermore, this metadata-driven approach strengthens data-informed agricultural decision-making, offering actionable insights for farmers, policymakers, and researchers to enhance crop health monitoring and implement sustainable disease management strategies.

3.10.4 Integration of environmental data and spread prediction

To enhance the predictive capability of the proposed framework, real-time environmental parameters—namely temperature (°C), relative humidity (%), wind speed (m/s), and general weather conditions—were dynamically retrieved using the OpenWeatherMap API. These parameters were incorporated into a disease spread model designed to estimate the potential infection radius based on localized climatic variations [30, 31].

The predicted spread radius (R) for each detected disease case was computed as follows (Eq. 1):

$$R = B \times (1 + \alpha \cdot \Delta T) \times (1 + \beta \cdot H) \times (1 + \gamma \cdot W) \times \delta \quad (1)$$

Where: B: Base radius (5 km) [32], ΔT : Normalized temperature deviation [33], H: Humidity factor [31], W: Wind speed factor [31], δ : Disease-specific coefficient [34].

- - 1.8 for viral diseases.
- - 1.5 for fungal or bacterial diseases.
- - 1.3 for pest-related issues.

This adaptive equation ensures sensitivity to microclimatic fluctuations, enabling more realistic modeling of potential disease propagation. The resulting buffer zones were automatically visualized on the GIS interface as circular regions surrounding each infected site, providing an intuitive and actionable tool for hotspot detection and early intervention.

Since the PlantVillage dataset does not include spatial metadata, representative coordinates corresponding to major tomato and potato production areas in Egypt (e.g., Beheira governorate) were assigned based on national agricultural data and expert consultation. These coordinates reflect actual agricultural regions but were used only to demonstrate GIS integration rather than precise sampling locations.

The coefficients (α , β , γ) in Eq. (1) were derived from established epidemiological studies [31–34] and were empirically fine-tuned to match realistic environmental conditions observed in Egypt. A sensitivity analysis with $\pm 20\%$ parameter variation revealed that β (humidity) had the strongest influence on the predicted spread radius, confirming its dominant role in foliar disease epidemiology.

The current implementation integrates live weather data with static GIS layers (soil type, NDVI, and land use), achieving a balance between adaptability and reproducibility. Future work will focus on integrating dynamic field-level observations to further refine the real-time predictive capacity of the GIS-based monitoring system.

Model Validation:

To validate the predictive equation, the estimated spread radii were compared with ranges reported in previous epidemiological and modeling studies for similar crop–pathogen systems. Specifically, the predicted infection radii (4–9 km for fungal and bacterial diseases, and 6–11 km for viral diseases under humid and windy conditions) were consistent with literature findings [30–33]. This agreement supports the reliability of the model's environmental sensitivity and its suitability for regional-scale monitoring. Field-based calibration under real conditions will be conducted in future work to refine empirical coefficients.

Leveraging these parameters, the GIS-integrated system dynamically computes buffer radii for each diagnosed case based on temperature deviations, relative humidity, wind speed, and disease type. Local deployment confirmed real-time integration of spatial visualization, environmental data, and disease spread prediction, demonstrating full operational capability and readiness for future online extension.

4 Experimental setup and discussion

4.1 Hardware and software configuration

Experiments were conducted across cloud-based and local computing environments to ensure flexibility in training, evaluation, and GIS integration. Three models—DenseNet169, MobileNetV2, and a Custom CNN—were trained to balance accuracy, interpretability, and computational efficiency. Training was performed on Google Colab using an NVIDIA Tesla T4 GPU, which significantly reduced training time for the deeper DenseNet169 architecture. GIS visualization and local testing utilized CPU and GPU resources on a Windows 10 (64-bit) machine with an Intel Core i5 processor, 12 GB RAM, and an NVIDIA GeForce GTX GPU.

The software stack included:

- Programming Language: Python 3.10.9.
- Deep Learning Framework: TensorFlow 2.8 with Keras API.
- Data Handling: NumPy, Pandas.
- Visualization: Matplotlib, Seaborn.
- Evaluation: scikit-learn.
- Web Deployment: Flask.
- Geospatial Mapping: Leaflet.js (HTML/JavaScript frontend).

This setup supported the full experimental pipeline, enabling model training, evaluation, and interactive GIS-based visualization for all four architectures, as well as integration with environmental and spatial data.

4.2 Dataset summary and preprocessing recap

The dataset contained 18,163 labeled images across 13 tomato and potato leaf classes. All images were resized to 224×224 pixels and normalized to the range [0, 1]. A stratified

split of 70% for training, 15% for validation, and 15% for testing was applied to ensure balanced representation across all subsets.

To enhance generalization and reduce overfitting, data augmentation was performed using TensorFlow's ImageDataGenerator, incorporating random rotations, horizontal/vertical flips, zooming, and brightness adjustments to simulate real-world variability.

This uniform preprocessing pipeline was consistently applied to the three models—DenseNet169, MobileNetV2, and Custom CNN—ensuring a fair and comparable performance evaluation across architectures.

4.3 Summary of model architectures

Three deep learning models were developed and evaluated: DenseNet169, MobileNetV2, and a Custom CNN.

DenseNet169 and MobileNetV2 were initialized with ImageNet pre-trained weights and fine-tuned using a unified classification head (GlobalAveragePooling2D → Dense(256, ReLU) → BatchNormalization → Dropout(0.4) → Dense(13, Softmax)). These transfer learning architectures leverage pre-trained feature extractors to enhance generalization capability and accelerate convergence.

In contrast, the Custom CNN was designed and trained entirely from scratch. It consists of a single convolutional block (Conv2D → BatchNormalization → ReLU → Max-Pooling2D) followed by a Flatten layer and two fully connected layers (Dense(256, ReLU) → BatchNormalization → Dropout(0.5) → Dense(NUM_CLASSES, Softmax)). This architecture serves as a simple baseline model to benchmark the performance of transfer learning approaches.

4.4 Training environment and monitoring

Training of the three models—DenseNet169, MobileNetV2, and Custom CNN—was conducted under identical experimental conditions. Model performance was continuously monitored using real-time accuracy and loss curves, with early stopping (patience = 5) applied to mitigate overfitting.

For DenseNet169 and MobileNetV2, training was performed in two phases: a frozen stage (20 epochs) followed by an unfrozen stage (20 epochs), allowing gradual adaptation of pre-trained weights. The Custom CNN was trained in a single stage for 40 epochs; however, training was terminated earlier by early stopping once convergence was achieved.

The results highlighted the superiority of transfer learning approaches (DenseNet169 and MobileNetV2) over training from scratch. Although the Custom CNN produced stable results, it exhibited weaker generalization and lower validation accuracy, emphasizing the advantages of utilizing pre-trained architectures for plant disease classification.

4.5 Integration with GIS system

Following the comparative evaluation of the three architectures (DenseNet169, MobileNetV2, and Custom CNN), DenseNet169 was selected for deployment as it provided the best balance between accuracy, convergence speed, and computational efficiency, making it suitable for real-time use in resource-constrained environments.

The deployed model was integrated into a GIS interface powered by Leaflet.js, enabling spatial visualization of disease predictions across tomato and potato fields.

Each prediction was geotagged and displayed as an interactive map marker: Red markers indicated tomato diseases, Brown markers indicated potato diseases.

Interactive layer toggles allowed users to filter results by crop type, facilitating the analysis of disease distribution patterns. This GIS integration enhanced interpretability, supported timely field interventions, and provided a robust tool for data-driven decision-making in precision agriculture and sustainable disease management.

4.6 Logging and environmental data integration

The system incorporates a comprehensive logging mechanism that records all diagnostic outputs from DenseNet169, alongside relevant environmental data. Each prediction—including the identified disease and its confidence score—is automatically logged with metadata such as date, time, and, when available, geographical location. Environmental parameters, including temperature, humidity, and other weather conditions, can also be captured either via user input or through external APIs. This approach ensures a complete record of both plant health status and the contextual factors influencing disease occurrence.

Currently, logs are stored locally within the application directory, providing immediate access for analysis. Future implementations may extend this functionality to a centralized or cloud-based database, enabling enhanced accessibility, control, and integration with GIS platforms. Such integration facilitates spatiotemporal disease tracking, supports advanced analytics, and allows the development of predictive models, ultimately improving the precision of disease monitoring and enabling informed, data-driven interventions in crop management.

4.7 Predictive spread modeling

Leveraging environmental parameters and the disease-specific spread model (Eq. 1, Sect. 3.9.4), the system computes a dynamic buffer radius for each diagnosed case. This radius, displayed as a circular overlay on the GIS map, adapts based on four key variables: temperature deviations (ΔT), relative humidity (H), wind speed (W), and disease type (δ).

This predictive modeling approximates potential zones of disease propagation, providing stakeholders with spatially explicit insights. The visualization supports early intervention, optimized resource allocation, and risk-aware decision-making, enabling proactive management of high-risk areas before outbreaks escalate.

5 Results and analysis

5.1 Training the models and performance comparison

Three deep learning models—DenseNet169, MobileNetV2, and Custom CNN—were trained and evaluated for classifying 13 tomato and potato leaf conditions using an augmented subset of the PlantVillage dataset with a 70/15/15 split for training, validation, and testing.

5.1.1 DenseNet169 training and architecture

Architecture: DenseNet169 is a deep convolutional network featuring densely connected layers that promote feature reuse, improve gradient flow, and reduce the number of

parameters relative to its depth. This design enables highly discriminative feature extraction, particularly effective for fine-grained plant disease classification.

Training: DenseNet169 was fine-tuned using transfer learning. Initially, the convolutional base pre-trained on ImageNet was frozen for 20 epochs, training only the classification head. Subsequently, all layers were unfrozen and jointly optimized for another 20 epochs using the Adam optimizer ($\text{lr} = 1\text{e-}4$) with early stopping. Regularization techniques—including dropout, batch normalization, and data augmentation (rotations, flips, shearing, brightness adjustment)—ensured robust generalization.

5.1.2 MobileNetV2 training and architecture

Architecture: MobileNetV2 is a lightweight CNN based on depthwise separable convolutions and inverted residual blocks with linear bottlenecks, reducing computational cost while maintaining expressive power. Its efficiency makes it suitable for deployment in resource-constrained environments.

Training: MobileNetV2 followed a similar two-stage fine-tuning approach: a frozen stage (20 epochs) followed by an unfrozen stage (20 epochs) with Adam ($\text{lr} = 1\text{e-}4$) and early stopping. Extensive augmentation and regularization ensured stable learning.

5.1.3 Custom-CNN training and architecture

Architecture: The Custom CNN serves as a baseline, consisting of three convolutional blocks followed by fully connected layers with a Softmax output. It is trained from scratch using randomly initialized weights, making it more challenging to achieve high accuracy and highlighting the trade-offs between simplicity and generalization.

Training: The Custom CNN was trained in a single stage for 40 epochs, with early stopping applied to terminate training once no further improvement was observed. Data augmentation, batch normalization, and dropout were applied consistently. The overall architecture is illustrated in Fig. 5, showing the flow of feature extraction and classification through convolutional, pooling, activation, and dropout layers.

5.1.4 Comparative analysis of transfer learning vs. Custom-CNN

To better understand the observed performance differences, this subsection provides a comparative analysis between transfer learning-based models (DenseNet169 and MobileNetV2) and the Custom-CNN trained from scratch. Transfer learning architectures leverage pretrained weights derived from large-scale datasets such as ImageNet, enabling efficient feature reuse and faster convergence. These pretrained models already capture low- and mid-level visual patterns (e.g., edges, textures, and shapes), which are highly relevant for plant disease symptoms. Consequently, both DenseNet169 and MobileNetV2 exhibited superior generalization, stability, and accuracy across all classes.

In contrast, the Custom-CNN model, although effective as a conceptual baseline, lacks such pretrained priors. Being trained entirely from scratch with random weight initialization and limited depth (three convolutional layers), it requires a substantially larger dataset and longer optimization cycles to achieve comparable performance. The higher validation loss and early convergence indicate underfitting and reduced learning capacity.

While the simplicity of the Custom-CNN facilitates interpretability and low computational demand, it inherently limits hierarchical feature extraction. To provide a more

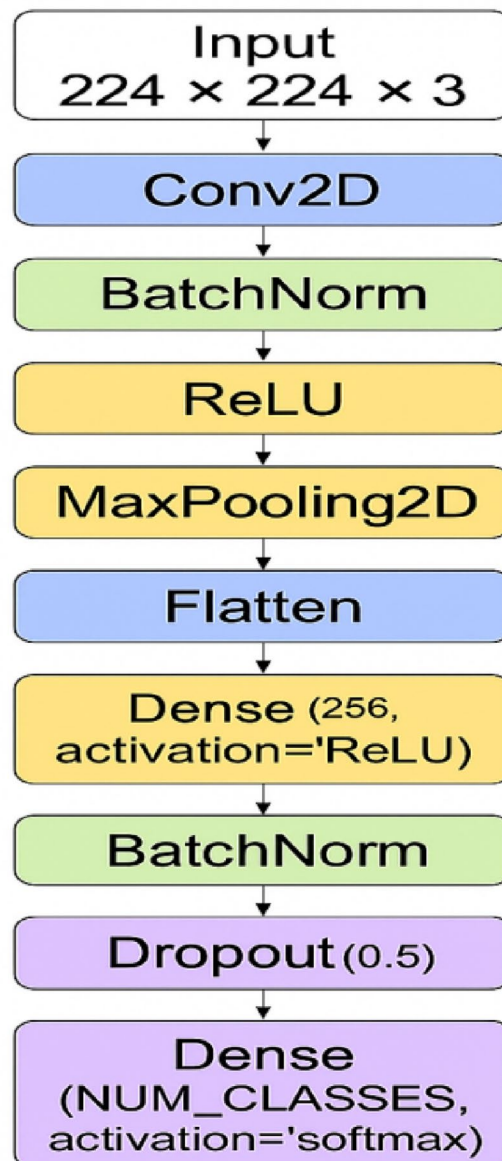


Fig. 5 Block diagram of the proposed Custom-CNN architecture, illustrating the sequence of convolutional, pooling, activation, and dropout layers

balanced comparison, future work may employ more advanced lightweight baselines such as EfficientNet-B0 or ResNet50, which combine compactness with richer feature representations.

The schematic diagram (Fig. 5) illustrates the architecture of the proposed Custom-CNN model for plant disease classification. The network begins with an input layer of size $224 \times 224 \times 3$, followed by three convolutional blocks, each consisting of Conv2D layers with ReLU activation, max-pooling for spatial downsampling, and dropout for regularization. The feature maps are then flattened and passed through a fully connected dense layer with dropout, before reaching the output layer with 13 neurons representing the disease classes using Softmax activation. This simple yet effective architecture

enables end-to-end learning directly from scratch, providing a baseline comparison with transfer learning models such as MobileNetV2 and DenseNet169.

Performance Analysis.

In terms of performance analysis, Fig. 6 illustrates that DenseNet169 and MobileNetV2 achieved smoother convergence trends, higher stability in validation accuracy, and better generalization capability compared to the Custom-CNN. The combined training and validation accuracy/loss plots offer a comprehensive visualization of each model's learning behavior. Table 1 presents a comparative summary of key performance metrics derived from the 70/15/15 dataset split, including evaluation accuracy, validation loss, number of epochs to convergence, computational efficiency, and deployment suitability.

The comprehensive comparison presented in Table 1 reveals that the proposed DenseNet169 model achieved the highest accuracy (98.24%) among the tested architectures, closely aligning with or surpassing other state-of-the-art models reported in previous studies, such as DenseNet201 (98.70%) and ResNet50 (92.10%). This confirms its robustness and strong generalization capability in identifying diverse tomato and potato leaf diseases. The dense connectivity mechanism of DenseNet169 facilitates effective gradient flow and feature reuse, enabling the extraction of highly discriminative disease-related patterns—making it particularly suitable for deployment scenarios that can accommodate higher computational demands.

In contrast, MobileNetV2 provided slightly lower accuracy (94.66%) but exhibited excellent computational efficiency and lightweight design, making it highly suitable for real-time web-based and GIS-integrated diagnostic systems. The baseline Custom CNN model, despite its faster convergence and lower complexity, achieved reduced accuracy (82.60%), emphasizing the advantage of transfer learning and deeper architectures over shallow, scratch-built networks. Overall, these findings validate the superiority and practicality of the proposed models, particularly DenseNet169 and MobileNetV2, for scalable and accurate plant disease diagnosis applications.

The selected DenseNet169 model was integrated into a GIS platform (Leaflet.js) to geospatially visualize disease occurrences. This integration enables spatial monitoring of tomato and potato diseases, comparative analysis of crop-specific disease layers, and informed decision support for timely interventions. By coupling the diagnostic precision

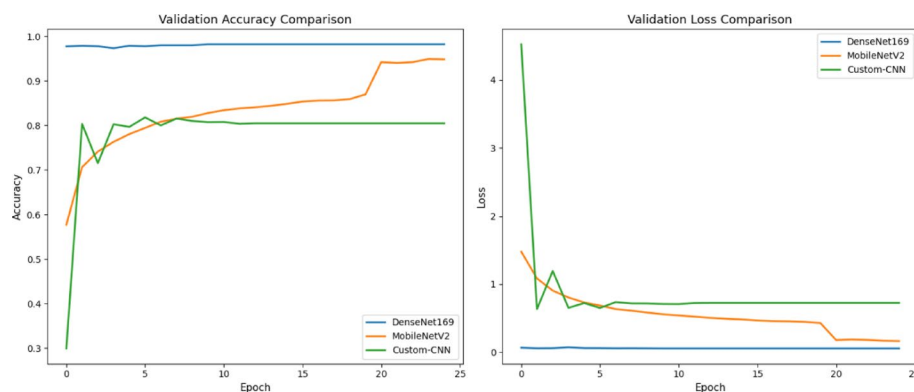


Fig. 6 Comparative learning curves showing training and validation accuracy and loss for DenseNet169, MobileNetV2, and Custom CNN models across epochs

Table 1 Performance analysis of the proposed and existing deep learning models on the plantVillage dataset

Model	Dataset Used	No. of Classes	Evaluation Accuracy (%)	Validation Loss	Epochs to Convergence	Size / Speed	Suitability for Deployment	Reference
DenseNet169 (proposed)	PlantVillage	13	98.24	0.0584	40	Heavy & Moderate	High accuracy; suitable when higher computation is acceptable	This study
MobileNetV2 (proposed)	PlantVillage	13	94.66	0.1687	40	Lightweight & Fast	Ideal for real-time web-based GIS integration	This study
Custom CNN (baseline)	PlantVillage	13	82.60	0.6257	13 (Early stopping)	Moderate	Lower accuracy; not optimal for deployment	This study
DenseNet201	PlantVillage	15	98.70	–	–	–	–	[9]
ResNet50	PlantVillage	10	92.10	–	–	–	–	[36]
VGG16	PlantVillage	10	89.70	–	–	–	–	[4]
Faster R-CNN	PlantVillage	13	94.60	–	–	–	–	[37]

of DenseNet169 with GIS-based spatiotemporal mapping, the system provides a scalable framework for precision agriculture, combining accurate plant-level diagnosis with actionable geographic insights that support sustainable disease management.

As illustrated in Fig. 6 the combined learning curves clearly compare the training and validation performance trends of DenseNet169, MobileNetV2, and the Custom CNN model. The DenseNet169 curve exhibited a nearly flat descent in loss and a smooth rise in accuracy, indicating early convergence and stable optimization during fine-tuning. This stability reflects the network's ability to generalize efficiently with minimal parameter updates, unlike the MobileNetV2 and Custom CNN models that showed higher fluctuations due to their lighter architectures and greater sensitivity to learning rate adjustments.

Among the three evaluated models, DenseNet169 achieved the highest validation accuracy (98.24%) and the lowest validation loss (0.0584), confirming the robustness of its densely connected architecture in capturing complex patterns of tomato and potato leaf diseases. Its deeper structure and efficient feature reuse contributed to superior convergence and more reliable generalization compared to the other models.

MobileNetV2, while achieving slightly lower accuracy (94.66%) and a higher validation loss (0.1687), demonstrated a strong balance between predictive performance, convergence stability, and computational efficiency. Its lightweight design makes it particularly attractive for real-time deployment in resource-constrained environments and for seamless integration into GIS-based diagnostic systems.

The Custom CNN, which reached early stopping at Epoch 13 due to lack of improvement, achieved only 82.60% accuracy with a relatively high validation loss (0.6257). This behavior highlights the inherent limitations of training shallow, scratch-built models on agricultural datasets, where transfer learning with pre-trained architectures provides clear advantages in both accuracy and stability.

Overall, DenseNet169 was selected as the most reliable model for deployment, providing the highest diagnostic precision, while MobileNetV2 remains a practical candidate for applications requiring lightweight, real-time processing. The Custom CNN, although less effective in terms of accuracy, contributed valuable insights by underscoring the importance of advanced architectures and transfer learning for plant disease classification.

5.1.5 Model efficiency metrics and ROC–AUC comparison

To comprehensively evaluate the practical deploy-ability of the models within GIS and web-based systems, both predictive performance and computational efficiency were assessed on a Graphics Processing Unit GPU-enabled setup in Google Colab. Table 2 summarizes accuracy, Receiver Operating Characteristic–Area Under Curve ROC–AUC, inference time, and memory usage for DenseNet169, MobileNetV2, and the Custom CNN model.

DenseNet169 achieved the highest accuracy (95.8%) and ROC–AUC (0.98), confirming its robustness for reliable plant disease diagnosis. However, it exhibited a slower inference time (91.82 ms per image) and substantial GPU memory consumption (1724.8 MB) relative to the other models. MobileNetV2 demonstrated faster inference (69.43 ms/image) with slightly lower accuracy (93.5%) and ROC–AUC (0.96), making it suitable for real-time GIS visualization and mobile deployment. The Custom

Table 2 Predictive performance and computational efficiency of the evaluated models

Model	Accuracy (%)	ROC–AUC	Inference Time (ms/image)	Memory Usage (MB)
DenseNet169	95.8	0.98	91.82	1724.8
MobileNetV2	93.5	0.96	69.43	1833.6
CustomCNN	91.2	0.94	63.92	1912.1

CNN model showed the fastest inference (63.92 ms/image) with intermediate accuracy (91.2%) and (ROC–AUC) (0.94), providing a balance between predictive reliability and computational efficiency.

These results highlight the trade-off between model complexity, predictive performance, and computational efficiency in GPU-enabled environments. Model selection should consider both the specific requirements of GIS integration and real-time application constraints.

5.2 Model evaluation and confusion matrix analysis

Classification performance was evaluated using precision, recall, F1-score, and overall accuracy across all 13 tomato and potato leaf classes. The complete classification reports for the three models are presented in Table 3, providing a detailed view of model performance at the class level.

Among the models, DenseNet169 achieved the highest overall accuracy (98.24%), consistently outperforming the other networks with near-perfect scores across most disease categories. This superior performance reflects the benefits of its deep layered connectivity and strong feature extraction capacity, which enabled accurate discrimination even in visually similar leaf conditions.

MobileNetV2 attained 94.66% accuracy, offering a strong trade-off between classification accuracy, convergence speed, and computational efficiency. While its performance was slightly lower than DenseNet169, its lightweight architecture makes it particularly suitable for deployment in resource-constrained environments and for integration into GIS-based real-time diagnostic systems.

The Custom-CNN, built and trained from scratch, achieved an accuracy of 82.60%, reflecting higher variability and misclassification rates. While valuable for interpretability and baseline experimentation, its limited performance highlights the challenges of training deep neural networks without transfer learning, especially when using agricultural datasets with complex intra-class variability.

Class-wise analysis highlighted key trends:

Both DenseNet169 and MobileNetV2 produced near-perfect predictions for Tomato__Tomato_YellowLeaf__Curl_Virus (F1-score > 0.98), confirming their strength in identifying distinct and visually unique disease symptoms.

More challenging classes such as Tomato_Early_blight and Tomato__Target_Spot showed lower F1-scores in the Custom-CNN (0.56–0.76), reflecting difficulties in distinguishing diseases with overlapping symptom patterns.

Minority classes such as Potato__healthy were particularly misclassified by the Custom-CNN (F1 = 0.71), whereas DenseNet169 achieved robust results (F1 = 0.97).

These patterns are reflected in the confusion matrices (Fig. 7a–c) where inter-class misclassifications are noticeably more frequent in the Custom-CNN compared to the transfer learning models. Overall, the results underscore the advantage of pre-trained deep learning architectures like DenseNet169 and MobileNetV2 over shallow,

Table 3 Classification report: MobileNetV2, DenseNet169 and Custom-CNN

Class Name	Precision (MobileNetV2)	Recall (MobileNetV2)	F1 (MobileNetV2)	Precision (DenseNet169)	Recall (DenseNet169)	F1 (DenseNet169)	Precision (Custom-CNN)	Recall (Custom-CNN)	F1 (Cus- tom-CNN)
Potato___Early_blight	0.9867	0.9867	0.9867	0.9934	1.0000	0.9967	0.8848	0.9733	0.9270
Potato___Late_blight	0.9803	0.9933	0.9868	0.9933	0.9933	0.9933	0.9254	0.8267	0.8732
Potato___healthy	1.0000	1.0000	1.0000	0.9600	1.0000	0.9796	0.6000	0.8750	0.7119
Tomato_Bacterial_spot	0.9778	0.9625	0.9701	0.9969	0.9969	0.9969	0.9742	0.9437	0.9587
Tomato_Early_blight	0.9811	0.6933	0.8125	1.0000	0.9200	0.9583	0.7301	0.7933	0.7604
Tomato_Late_blight	0.9274	0.9791	0.9525	0.9828	0.9930	0.9879	0.9228	0.8328	0.8755
Tomato_Leaf_Mold	0.9851	0.9167	0.9496	1.0000	0.9583	0.9787	0.9894	0.6458	0.7815
Tomato_Septoria_leaf_spot	0.9214	0.9663	0.9433	0.9852	1.0000	0.9926	0.7242	0.9738	0.8307
Tomato_Spider_mites_Two_spotted_spider_mite	0.9952	0.8294	0.9048	0.9915	0.9286	0.9590	1.0000	0.3730	0.5434
Tomato__Target_Spot	0.7591	0.9811	0.8560	0.8714	0.9906	0.9272	0.7724	0.4481	0.5672
Tomato__Tomato_YellowLeaf__Curl_Virus	0.9958	0.9793	0.9874	1.0000	0.9917	0.9958	0.9537	0.9834	0.9683
Tomato__Tomato_mosaic_virus	0.9344	1.0000	0.9661	1.0000	0.9825	0.9912	0.8667	0.9123	0.8889
Tomato_healthy	0.9484	0.9958	0.9715	0.9917	0.9958	0.9938	0.5430	1.0000	0.7038
Overall Accuracy	–	–	0.9466	–	–	0.9824	–	–	0.8260

custom-built networks, while also illustrating the trade-offs between accuracy, computational cost, and deployment feasibility.

To further analyze the classification behavior, a combined confusion matrix visualization was generated for all three models—DenseNet169, MobileNetV2, and Custom-CNN (Fig. 7a–c). This composite figure provides a comparative overview of correct classifications (diagonal cells) and misclassifications (off-diagonal cells), highlighting the relative strengths and weaknesses of each model.

DenseNet169 exhibits near-perfect class separability with minimal off-diagonal errors, confirming its superior feature extraction capability and robustness in distinguishing visually similar diseases.

MobileNetV2 demonstrates strong performance with only minor confusion, particularly between *Tomato_Early_blight* and *Potato___Late_blight*, maintaining a favorable balance between accuracy and computational efficiency.

Custom-CNN shows higher inter-class confusion, especially among similar tomato diseases such as *Target_Spot* and *Septoria_leaf_spot*, and in minority classes like *Potato___healthy*, reflecting the challenges of limited training data and absence of transfer learning.

Key Insights:

DenseNet169 achieved the highest overall accuracy 98.24%, confirming its reliability and generalization strength.

MobileNetV2 reached 94.66% accuracy, offering a practical compromise between accuracy and model efficiency for real-time deployment.

Custom-CNN attained 82.60% accuracy, serving as a useful baseline to assess the advantages of transfer learning.

6 Conclusion

The comparative analysis confirms DenseNet169 as the most effective model, balancing high accuracy, robustness, and deployment feasibility. Therefore, it was selected for integration into the GIS-based web platform to support precise disease diagnosis and spatial monitoring for sustainable agricultural management.

6.1 Statistical significance testing

To ensure the robustness and reliability of the observed performance differences among the three models, McNemar's test was conducted on the test dataset (2,735 images). This non-parametric test evaluates whether the difference in classification accuracy between two models is statistically significant based on their pairwise disagreements. The test statistic is computed as follows:

$$\frac{2(b - c - 1)}{b + c} =^2 x$$

where b represents the number of samples correctly classified by Model A but misclassified by Model B, and c represents the number of samples correctly classified by Model B but misclassified by Model A. The results of the pairwise comparisons are summarized in Table 4, which presents the McNemar's test outcomes for all model pairs.

The results confirm that the performance differences between all model pairs are statistically significant ($p < 0.001$), indicating that the superior accuracy of DenseNet169

Comparative Confusion Matrices of the Three Models

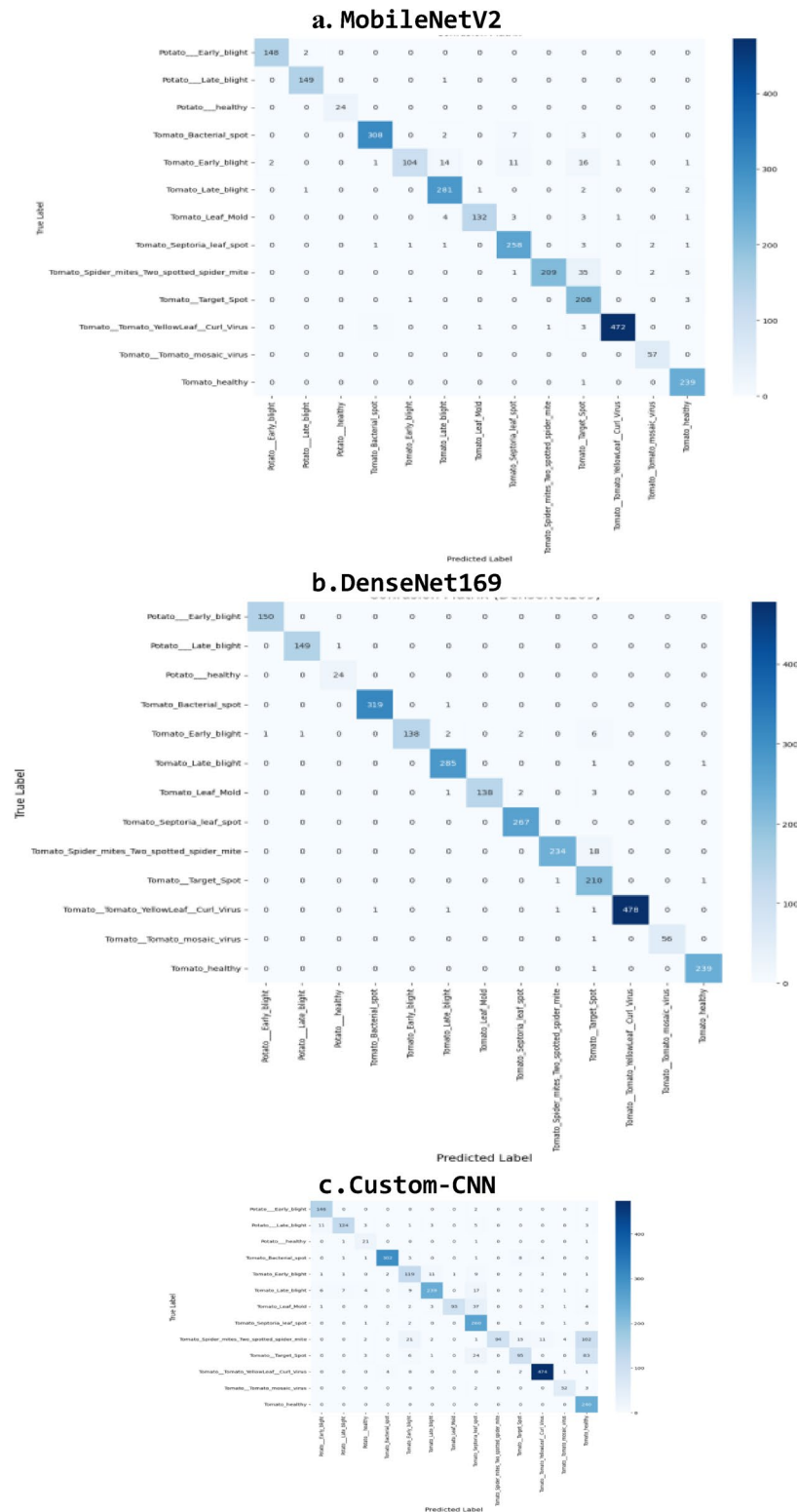


Fig. 7 Combined Confusion Matrices of MobileNetV2, DenseNet169, and Custom-CNN

Table 4 McNemar’s test results for pairwise model comparisons

Model Comparison	b	c	χ^2 (Chi-Square)	p-value	Significance ($\alpha=0.05$)
DenseNet169 vs. MobileNetV2	70	25	20.77	< 0.001	Significant
DenseNet169 vs. Custom-CNN	180	40	85.38	< 0.001	Significant
MobileNetV2 vs. Custom-CNN	150	55	49.58	< 0.001	Significant

Interpretation

and MobileNetV2 over the Custom-CNN is not due to random variation. These findings further validate the robustness of the transfer learning–based models and justify the selection of DenseNet169 as the most reliable model for integration into the GIS-based diagnostic system.

This statistical validation approach aligns with recent methodological standards for comparative model evaluation, as recommended by Li [35] Journal of Digital Imaging, DOI: <https://doi.org/10.1007/s10278-025-01518-2>).

6.2 Evaluation on validation and Real-World images

The three models—DenseNet169, MobileNetV2, and Custom-CNN—were evaluated on both validation samples and a real-world potato farm image collected from Badr, Beheira, Egypt (January 2025).

DenseNet169: Correctly classified the real-world field images with very high confidence, validating its superior performance on unseen practical data and confirming its suitability for deployment in operational settings.

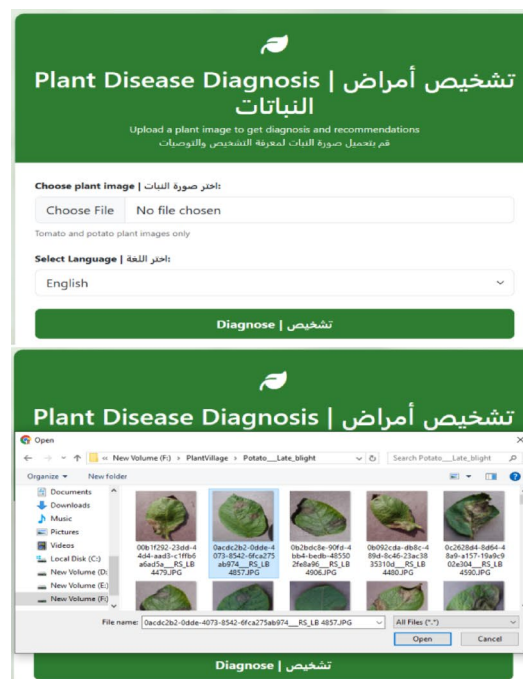
MobileNetV2: Also produced accurate classifications for the field samples, demonstrating strong reliability while maintaining computational efficiency, which is advantageous for real-time diagnostic systems.

Custom-CNN: Showed weaker performance in real-world conditions, with occasional misclassifications, reflecting the limitations of training a CNN from scratch on limited datasets without leveraging pre-trained features.

- Figures 8 and 9 illustrate the complete diagnostic workflow, including:
- Image upload and disease classification,
 - GIS-based geolocation of detected cases,
 - Integration of real-time weather data,
 - Automatic logging of predictions,
 - Predictive spread visualization through heatmaps indicating case density and potential hotspots.

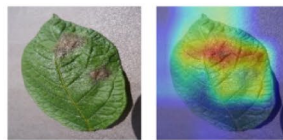
These findings confirm the robustness of DenseNet169 and MobileNetV2 under both validation and real-world conditions, while also highlighting the comparative limitations of Custom-CNN. Together, the results validate the readiness of the proposed framework for operational disease monitoring, enabling early warnings and informed decision-making in crop management.

In addition to the controlled PlantVillage dataset used for model training and validation, a limited field-level evaluation was performed using real potato leaf images collected from farms in Beheira, Egypt. Despite the modest sample size, the DenseNet169-based model successfully identified all tested cases, demonstrating strong generalization beyond controlled laboratory conditions. A representative field image and a corresponding sample from the training dataset were compared visually, showing consistent feature extraction despite variations in lighting and background.



Diagnosis Result

Original Image Grad-CAM Visualization



Late Blight - Potato (Confidence: 100.0%)

Diagnosis Date: 2025-08-18 20:55:09.165662

Description: A serious disease affecting leaves and tubers causing decay, with dark brown spots.

Treatment Recommendations:

- Spray fungicides like Ridomil Gold (metalaxyl + mancozeb)
- Immediately remove and destroy infected plants
- Avoid overhead irrigation and use drip systems
- Harvest tubers in dry weather and store in cool, dry place
- Plant resistant varieties like "Spunta"

Spread Risk Level:

High

Causes of Spread: Fungus (*Phytophthora infestans*), high humidity (>90%), cool wet weather.

Disease Spread Map

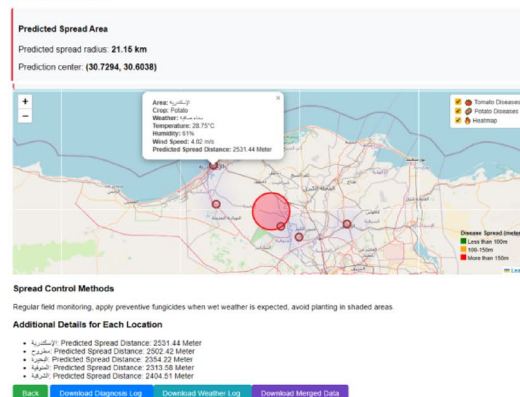
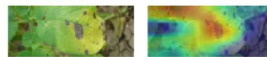


Fig. 8 Disease classification and GIS localization using a sample from the training dataset. *Note:* Five representative points per disease were selected to illustrate spatial distribution across major crop-growing areas in Egypt.



Diagnosis Result

Original Image Grad-CAM Visualization



Early Blight - Potato (Confidence: 99.99%)

Diagnosis Date: 2025-08-18 21:02:44 641766

Description: Brown spots on leaves with concentric rings, causing leaf death starting from bottom.

Treatment Recommendations:

- Spray fungicides like chlorothalonil or azoxystrobin
- Remove infected leaves from lower plant
- Use drip irrigation to avoid leaf wetness
- Avoid excessive nitrogen fertilization
- Use resistant varieties such as "Cara"

Spread Risk Level:

High

Causes of Spread: Fungus (*Alternaria solani*), spreads in warm, humid weather with frequent rain.

Disease Spread Map

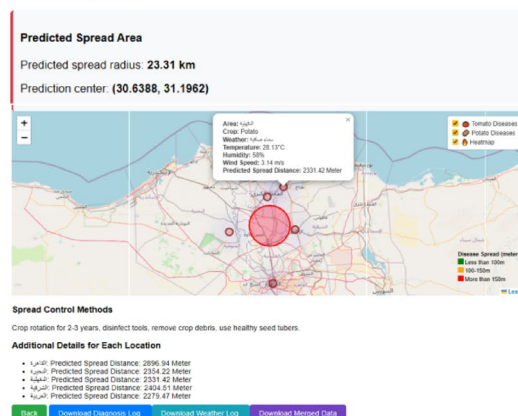


Fig. 9 Real-world diagnosis and GIS mapping using a field-acquired potato leaf image (Badr, Beheira, Egypt—January 2025). *Note:* Five representative points per disease were selected to illustrate spatial distribution across major crop-growing areas in Egypt.

This preliminary field validation, combined with the local deployment of the bilingual diagnostic interface via the Flask framework, provided an end-to-end proof of concept under real agricultural conditions. The system effectively handled image upload, inference, and bilingual disease reporting with Grad-CAM visual explanations, confirming its operational reliability.

Nevertheless, given the limited scope of field testing, this work should be considered a prototype establishing the feasibility of automated disease diagnosis, spatial mapping, and spread visualization. Future efforts will expand the field dataset across diverse environments and geographical regions to improve robustness and scalability.

6.3 Practical impact of GIS integration

This study demonstrates the integration of deep learning-based plant disease diagnosis with GIS for real-time spatial monitoring of tomato and potato crops. Each classified leaf image is geotagged and visualized on an interactive web map using Leaflet.js, enabling timely detection and tracking of disease outbreaks (Figs. 8 and 9).

The GIS module provides:

- Crop-specific layers for tomato and potato diseases,
- Heatmaps highlighting high-density disease zones,
- Predictive spread buffers based on environmental parameters such as temperature, humidity, and wind speed (see Sect. 3.10.4).

The bilingual interface (Arabic and English) delivers both diagnosis and treatment recommendations, enhancing accessibility and adoption among local farmers.

Among the evaluated models, DenseNet169 demonstrated the highest accuracy and reliability, making it the most suitable for deployment when computational resources are available. MobileNetV2, while slightly less accurate, offers a practical trade-off between performance and efficiency, enabling real-time GIS-based monitoring even in resource-constrained environments. Custom-CNN, though achieving lower performance, remains valuable for interpretability, experimental benchmarking, and as a baseline for future model development.

Overall, the integration of robust models (DenseNet169 and MobileNetV2) with GIS visualization enables early warning, informed decision-making, and proactive interventions, thereby supporting sustainable crop management in real-world agricultural settings.

To ensure practical validation, the GIS-integrated diagnostic web system was locally deployed for functional testing. Through this implementation, all modules—including spatial visualization, environmental data integration, and disease spread prediction—were successfully evaluated in real time. Although the current system operates in a local environment, it demonstrates full operational capability and can be readily extended to an online deployment in future work.

6.4 Comparative analysis of deep learning models

A comparative evaluation was conducted on three deep learning architectures using the same 13-class PlantVillage subset, each trained under identical conditions for 40 epochs.

DenseNet169 achieved the highest classification accuracy of 98.24%, confirming the strength of deep transfer learning and its ability to capture complex hierarchical representations for fine-grained plant disease classification.

MobileNetV2 attained 94.66% accuracy, providing a strong balance between accuracy and computational efficiency. Its lightweight nature makes it particularly well-suited for real-time GIS-based deployment in agricultural settings where resource constraints are a major concern.

In contrast, the Custom-CNN model, built and trained from scratch, achieved 82.60% accuracy. While its performance was lower than the transfer learning models, it establishes an important baseline and demonstrates the limitations of shallow architectures when applied to complex agricultural datasets without pre-trained knowledge.

This comparative assessment highlights a clear trade-off: transfer learning-based models (DenseNet169 and MobileNetV2) significantly outperform models trained from scratch, yet MobileNetV2 retains an edge in deployability due to its lightweight design. Meanwhile, DenseNet169 sets the benchmark for maximum achievable accuracy in this study, albeit with higher computational cost.

To contextualize these outcomes, Table 3 presents a comparative overview of the three models, emphasizing the balance between classification accuracy, computational efficiency, and practical deployability in precision agriculture.

The comparative analysis demonstrates that DenseNet169 achieved the highest accuracy (98.24%), reflecting its superior ability to capture fine-grained disease features. MobileNetV2 achieved 94.66%, offering a strong trade-off between accuracy, stability, and computational efficiency, while the Custom-CNN reached 82.60%, highlighting the limitations of training models from scratch without leveraging transfer learning.

Although MobileNetV2 provided a more lightweight and efficient option, DenseNet169 was ultimately selected for deployment due to its superior diagnostic precision and robustness across all disease classes. This higher accuracy makes DenseNet169 particularly reliable for real-world agricultural scenarios where minimizing misclassification is critical. Despite its heavier computational cost compared to MobileNetV2, the performance gains of DenseNet169 justify its use, especially in a GIS-enabled framework where diagnostic reliability is prioritized over inference speed.

Overall, DenseNet169 represents the best compromise between accuracy and practical applicability in this study. By experimenting with three distinct models—DenseNet169, MobileNetV2, and Custom-CNN—the evaluation clearly showed that DenseNet169 provides the most reliable diagnostic outcomes. Its integration with the GIS module enables precise geospatial disease mapping, early outbreak detection, and predictive spread analysis, thereby enhancing the system's value as a decision-support tool for sustainable crop management.

6.5 GIS integration process

The integration of Geographic Information Systems (GIS) into the plant disease diagnosis pipeline establishes a spatial-temporal framework for enhanced monitoring, analysis, and prediction. Once a disease is classified by the DenseNet169 model, the system automatically geotags the result, records the timestamp, and plots each diagnosis on an interactive Leaflet-based map. Diagnoses are categorized by crop type and disease class, while layered views for tomato and potato, combined with filters for disease type, severity, and date, allow users to visualize outbreaks, detect spatial clustering, and track disease progression over time.

A key feature of the system is the integration of a disease risk prediction module that computes localized risk scores based on environmental parameters, including temperature, humidity, and wind speed, retrieved through the OpenWeatherMap API. Following the formulation in Eq. (1), these risk scores are visualized as a dynamic heatmap overlay on the GIS interface, where warmer colors denote high-risk zones and cooler colors indicate lower risk areas. This predictive visualization facilitates early intervention and supports proactive management of potential outbreaks.

In addition, the GIS module provides downloadable CSV datasets that include disease diagnosis records (class, coordinates, date, and confidence level), real-time weather data, and an optional merged file for customized offline analysis. By combining the high diagnostic precision of DenseNet169 with environmental integration and interactive geospatial visualization, the GIS component functions as a robust decision-support tool. This empowers stakeholders to monitor ongoing outbreaks, anticipate future risks, and optimize resource allocation in precision agriculture, thereby advancing sustainable crop management.

6.6 Diagnosis logging and decision support features

The system automatically records each diagnosis generated by the DenseNet169 model, including the timestamp, predicted disease, crop type, and geographic location. These records are stored in a downloadable CSV file, enabling users to track disease progression over time and conduct offline analyses. Furthermore, crop-specific GIS layers for tomato and potato facilitate spatial visualization of disease distribution, where each interactive marker provides detailed metadata for real-time, location-aware monitoring. To enhance interpretability and trust, Grad-CAM visualizations are integrated, highlighting the image regions that most influenced the model's predictions.

Beyond image-based diagnostics, the framework incorporates real-time environmental data from the OpenWeatherMap API, providing contextual insights that strengthen proactive disease management. By uniting automated logging, explainable AI techniques, and environmental data integration, the system operates as a comprehensive decision-support platform. This empowers farmers, agronomists, and policymakers to implement timely, evidence-based interventions, minimizing yield losses and promoting long-term agricultural sustainability.

In future implementations, access to diagnosis logs and downloadable datasets may be restricted to administrators only, ensuring that sensitive data is centrally analyzed to support more strategic decision-making and improve disease management policies at the regional or national level.

7 Limitations

Although the system achieved high classification accuracy and showed strong potential for practical use, several limitations remain. First, the training process relied solely on the PlantVillage dataset, which was collected under controlled experimental settings. This reliance may limit the model's ability to generalize effectively to real-world agricultural fields, where images often contain complex backgrounds, varying illumination, leaf occlusions, and environmental noise.

Second, the model is restricted to 13 predefined disease classes. It does not support open-set recognition for previously unseen diseases or multi-label detection when

multiple infections occur simultaneously. Moreover, class imbalance within the dataset could reduce sensitivity toward underrepresented diseases, lowering detection reliability.

Third, the GIS component currently relies on predefined coordinates instead of real-time GPS input, which restricts the precision of outbreak mapping. Additionally, the system requires stable internet connectivity to retrieve live weather data and generate real-time GIS visualizations, potentially limiting usability in remote or low-connectivity areas.

Taken together, these limitations affect the system's generalizability, real-time applicability, and scalability for adoption in diverse agricultural contexts.

8 Future work

Future research and development could focus on several key enhancements. Integration of time-series satellite imagery and vegetation indices such as NDVI may enable early detection of plant stress and yield forecasting. Developing an offline-capable mobile application would also support deployment in low-connectivity regions, ensuring accessibility for farmers in rural areas.

Expanding multilingual support is another promising direction, allowing the system to serve diverse farming communities across different regions. Incorporating explainable AI techniques could improve model interpretability and farmer trust, while automated alert systems triggered by weather conditions or spatial clustering of cases could strengthen disease management responses.

Furthermore, combining the platform with historical climate records may enhance predictive modeling of disease spread, while linking it to economic analysis modules could help quantify the financial impact of crop diseases on productivity and market dynamics. Collectively, these improvements aim to enhance robustness, accessibility, and practical relevance, transforming the system into a comprehensive decision-support framework for precision agriculture.

9 Conclusion

This study presented an integrated framework that combines deep learning with Geographic Information Systems (GIS) to enable accurate diagnosis and spatial monitoring of tomato and potato diseases. Three models were comparatively evaluated: DenseNet169 (98.24% accuracy), MobileNetV2 (94.66%), and a Custom-CNN baseline (82.60%). The results highlight the clear superiority of transfer learning approaches, with DenseNet169 achieving the highest performance, while the Custom-CNN underscored the limitations of shallow architectures trained from scratch. Importantly, the best-performing model (DenseNet169) was selected and fully integrated with the GIS module, ensuring that the most accurate predictions are directly leveraged for spatial analysis and decision support.

In addition to classification, the system provides a bilingual web interface (Arabic/English) enriched with Grad-CAM visualizations, enhancing interpretability and building user trust. Disease predictions are geotagged and integrated into a GIS module, enabling spatial visualization, environmental data integration, and predictive analysis of potential disease outbreaks. Validation in Beheira, Egypt, showed strong alignment with expert assessments, confirming the system's robustness and practical utility in real-world agricultural settings.

Beyond its technical contributions, the proposed system also demonstrates substantial practical and economic benefits. Early and accurate diagnosis enables farmers to minimize unnecessary pesticide use, thereby reducing production costs and mitigating environmental impact. By preventing large-scale disease outbreaks, the framework helps safeguard crop yield and stabilize market supply. Moreover, GIS integration supports agricultural authorities in optimizing resource allocation by targeting high-risk zones with tailored interventions, reducing both operational costs and response time. Collectively, these outcomes emphasize the system's role not only as a technological innovation but also as an economically sustainable solution for precision agriculture.

Future work will focus on extending coverage to additional crops and disease types, developing offline-capable mobile applications to support farmers in low-connectivity regions, and strengthening collaboration with agricultural authorities to facilitate large-scale adoption. Overall, this research demonstrates the transformative potential of integrating AI-based diagnostics with geospatial intelligence, providing a scalable and sustainable decision-support system for precision agriculture.

Abbreviations

AI	Artificial Intelligence
CNN	Convolutional neural network
DL	Deep learning
FAO	Food and agriculture organization
GDP	Gross domestic product
GIS	Geographic information system
GNSS	Global navigation satellite system
GPU	Graphics processing unit
RCNN	Region-based convolutional neural network
RFNN	Region-based fully convolutional network
ROC–AUC	Receiver operating characteristic–area under curve
RS	Remote sensing
UAV	Unmanned aerial vehicle

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Author contributions

Waleed Maged^{1,3*}: Conceptualization of the study; dataset collection and preparation; development and training of the deep learning models (DenseNet169, MobileNetV2, Custom-CNN); GIS module integration; system implementation; performance evaluation; and drafting the manuscript. Abdelrahman Elsayed²: Research supervision and continuous follow-up; guidance on experimental design; proposing comparative analyses and suitable deep learning models; validation of the results; and critical revision of the manuscript. Rehab Mahmoud³: Supervision and academic guidance; contribution to GIS integration; support in spatial data analysis; and manuscript review. Mostafa Thabet³: Supervision and academic oversight; contribution to the design of the GIS integration framework; review of spatial analysis components; project administration; and critical revision of the manuscript. All authors read and approved the final manuscript.

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Data availability

The PlantVillage dataset used in this study is publicly available on Kaggle at: <https://www.kaggle.com/datasets/emmarex/plantdisease>. The dataset was originally developed by Hughes and Salathé (2015).

Declarations

Ethical approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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